

Applied Mechanistic Interpretability Methods

Reading is not steering: what a probe cannot tell you
about a steering vector

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Abstract

A steering vector is a direction added to a language model's residual stream, and the natural assumption behind one is that a direction which reads a concept well will also steer it. I test that assumption on GPT-2 small across six binary concept families, twelve layers and four sources of the direction, after one exploratory pass through activation patching, probing, steering and sparse autoencoders. A training-free predictor — the model's own unembedding applied at each layer — ranks the families in the order their steering effects arrive, while a trained probe saturates at 1.00 in every layer of two of them and predicts nothing; on one family the concept is perfectly readable in all twelve layers and steering moves the answer by at most +0.014. The correlation the anchor paper reports is real and inflated at once, because the dose it never states tracks its own predictor at +0.76 to +0.91: holding the dose fixed inverts the ranking of the families, and read at an equal side-effect budget instead the predicted order survives with one adjacent swap, with the layer and the dose mattering far more than where the direction was derived from. What each of these measurements is worth therefore depends on a setting chosen before it: at what dose, and compared at equal what.

Use of AI. This project was done with an AI coding assistant: literature search, notebook code, figures, drafts of this text, and discussion throughout. I chose the question, designed the experiments and did the interpretation. The working method is part of the deliverable rather than hidden behind it: the rule file, the context directory with project state and long-lived notes, and the plan agreed before each block of work are in the repository and can be inspected in full.

Path in the repository	What is in it
CLAUDE.md	The rules the collaboration runs under: what I decide, what is delegated, how a finding gets documented, and that a plan I did not approve is not a licence to start.
docs/agentcontext/PROJECT.md	Project state and the list of decisions, each with the reason it was taken.
docs/agentcontext/DETAILS.md	Long-lived notes, one section per method, model or finding; the larger ones live beside it in details/.
docs/agentcontext/plans/	The plan agreed before each larger block of work, finished ones moved to plans/done/.
docs/agentcontext/HISTORY.md	Dated chronology of what was done and why.
docs/learning/, docs/fahrplan/	My own explanations of the concepts and the roadmap I worked along — working documents, in German.



1 Introduction

Where this sits. Explainable AI is usually sorted along a few independent axes: whether the explanation is intrinsic to the model or added after the fact, whether it applies to any model or only to one kind, and whether it accounts for a single decision or for the behaviour as a whole [1]. The methods that carry the field in practice sit on the agnostic side and work from the outside: a local surrogate fitted around one prediction [2], attention maps, gradient attributions. They answer one question, which part of the input mattered for this output, and they treat the computation in between as given. Mechanistic interpretability asks what that computation is, stated in the model's own internal quantities. On the axes above it is post-hoc and as model-specific as a method can be: it needs the weights, the layers and the activations of one particular network. Everything in this report is measured inside GPT-2 small rather than around it.

What this report is. Four methods applied to one model: activation patching, probing, steering, and sparse autoencoders. I ran each of them once and small to learn what it measures and where it misleads (Section 3), and then took a single question far enough to produce numbers worth defending (Section 5): whether a direction that reads a concept well out of a layer also steers the model when it is written back in. No research question was fixed in advance; the exploration was the method. The report is therefore ordered the way the work happened: the instruments first, then one question chosen out of the literature, then one experiment on six concept families, twelve layers and four sources of a steering direction.

The stance. One sentence carries the rest. Each of these methods has a setting that must be chosen before anything can be measured (the corrupted baseline in patching, the control task in probing, the dose in steering, the sparsity level in a sparse autoencoder), and that setting co-determines the finding. Section 5 contains an instance of it: the same six concepts, the same model and the same directions change their ranking depending on whether the comparison is read at equal coefficient, at equal relative dose, or at equal side effect, and only the last of the three compares what a direction did at the same cost. A result from any of these methods is a statement about the model and about the instrument at once; the setting has to be named to keep the two apart.

How to read this. Every section below opens with a grey box saying what it does and why, so a reader can move from box to box and stop where the detail is needed. Passages marked with a vertical rule are the claims I am prepared to defend, as opposed to numbers I merely measured. Everything measured here is in the repository; the appendix gives the link, the tag, and what lies where.

2 Background: four ways to look inside

What and why. The four methods in this report read from or write into the same object, and they rest on the same assumption. This section defines both: the residual stream (Figure 1), the linear representation hypothesis in three sentences, and Table 1, which sets the instruments side by side by what each one measures, what it takes for granted, and where it breaks. A reader who knows the field needs only the last column of that table.

The workspace. A transformer of the GPT-2 family keeps one vector per token position — 768 numbers in GPT-2 small — and every block adds to it instead of replacing it [3]. Attention writes into it, the MLP writes into it, and what was written before stays. The state at layer ℓ is the embedding plus everything the first ℓ blocks contributed, and two consequences of that carry this report. First, the model's own output map is a linear read of that state, so it can be applied at any depth rather than only at the end: that is the logit lens [4], and A_{lin} in Section 5 is nothing more than how often this

early read already names the right answer. Second, because the contributions add, a single one can be measured, removed or amplified without rebuilding anything, and that is what makes all four methods possible in the first place. The stream is not scale-free, though. Measured on GPT-2 small, the mean residual norm grows from 76 at layer 0 to 644 at layer 10, a factor of about eight, so adding a vector of a given length is a different intervention at every depth. That is why Section 5 states every dose relative to its layer.

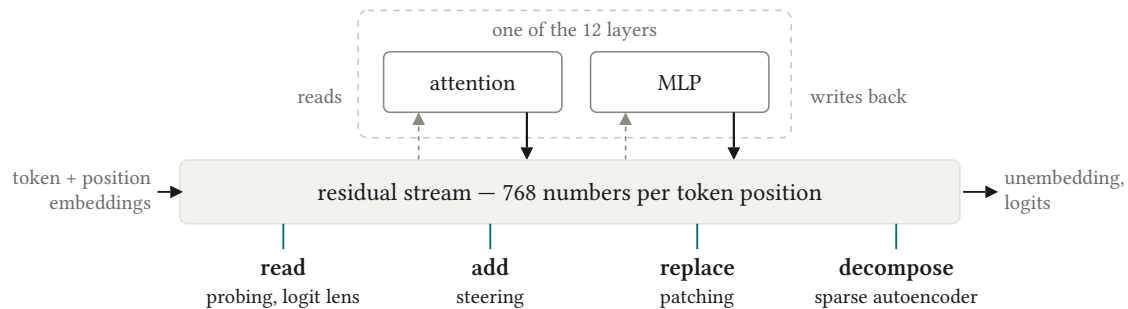


Figure 1: Own figure, schematic. Every block reads the whole stream and adds its result back into it, so the state at a layer is the embedding plus everything written into it so far. The four methods of this report are four operations on that one object.

The assumption. Three sentences hold the rest up. A concept is a direction in that space. Reading it is a projection: how much *day versus night* a state carries is a single number, its dot product with the direction. Writing it is an addition: put a multiple of the direction into the stream and the model continues as if the text had said so. This is the linear representation hypothesis [5], an assumption with enough support to build tools on, not a theorem. The four methods below do not test it; they use it.

The four instruments. Table 1 is the working definition of each method. The failures in its last column are not defects of the methods: each is the method’s own assumption showing up as a number, and each is answered by a control rather than by a better tool.

Method	What it measures	What it assumes	Where it breaks
Activation patching	How much of a behaviour one component causes: replace its activation with the one from a different run and read the change in logit difference [6].	That the two runs differ only in the intended way, and that a part can be swapped without disturbing what feeds it.	The ranking depends on the corrupted baseline and the metric as much as on the model [7], and later components repair what was removed, so necessity reads low.
Probing	Whether a concept can be read linearly out of one layer: the accuracy of a classifier trained on the activations [8].	That what is decodable is also what the model uses.	A probe with enough capacity also fits labels that carry no information; without a control task the accuracy means nothing [9], [10].
Steering	The causal effect of a direction: add a multiple of it to the stream and read what changes in the output [11].	That the direction is the concept, and that one coefficient means the same at every layer.	A random vector of the same norm buys much of the same damage, and with the dose unstated the effect of the direction cannot be told from the effect of its size.
Sparse autoencoder	A sparse decomposition of the state into features, scored by explained variance and by how many features fire [12].	That features are sparse and linear, and that a good reconstruction kept what matters.	Explained variance is blind to the concept: 93 to 99% of the state is reconstructed in Section 5 while up to 99% of the class separation sits in the discarded error.

Table 1: The four methods as this report uses them. The last column is each method’s own assumption, stated as the way it fails.

All four break in the same place: a setting of the instrument decides part of the answer. Section 3 is what that looked like when I ran each of them once.

3 Exploration: one station per method

What and why. Before choosing a question I ran every method once and small, one notebook per station: patching, steering, probing, sparse autoencoders, and one pass over a second model. These are single prompts and sample sizes around twenty; the aim was to learn what each instrument measures and where it misleads, not to find anything. Table 2 is the whole tour. Three of its lessons became the design of Section 5, and the final paragraph says why the experiment then ran on the smaller of the two models.

Station	What I asked	What came out	What it taught
Orientation 01_explore	Does the plumbing behave as documented?	Hook names verified against the library version; BOS carries a residual norm near 3100 against 250 elsewhere; the logit lens is unreadable before layer 6.	Drop position 0 from every mean. Lens probabilities show a direction, not a confidence.
Patching 02_patching	Which heads carry the induction behaviour?	Ablating the five canonical heads destroys 25% of the gap; patching them back restores 60%. No single head restores more than 0.13, and the one with the highest attention score restores 0.05.	Ablation measures necessity, patching sufficiency, and they disagree. The head with the highest attention score is not the one that matters most.
Steering 03_steering	What does a difference-of-means direction do, and at what price?	An S-curve that saturates before the loss does: the metric plateau sits where the loss is already rising. A random vector of the same size leaves the metric flat and costs most of the same loss.	State the dose relative to the layer's residual norm. A random control checks the size of an intervention, not the specificity of its direction.
Probing 04_probing	Is the probe direction different from the difference of means?	On 20 minimal pairs, no: $\cos(\hat{w}, d) \geq 0.95$ at every regularisation strength. The control task reaches 19.4 of 40 held out at training accuracy 1.00.	Minimal pairs leave the probe nothing to pick up except the concept, so they cannot separate reading from steering; that test needs heterogeneous data.
SAE 06_sae	Is a sparse feature a cleaner lever than a difference of means?	At equal dose it looks weaker; read at equal effect , both interpolated to +2.00 logits, it is a tie of 0.069 nats against 0.077. The opposite feature costs five times as much.	Compare at equal effect rather than at equal dose. The SAE feature is derived differently; it is not a better lever.
Second model 05_gemma	Does this depend on the quirks of one small model?	Gemma-2-2B matches its reference implementation to max $ \Delta = 0.375$ at a logit scale of 28.6; its BOS sink is in every layer, and the cache costs about 4 MB per token.	The BOS sink and the layer-relative dose are not GPT-2 artifacts. A second model costs memory, not correctness.

Table 2: The exploration tour, one row per notebook in notebooks/. The numbers are orientation at $n \approx 20$, not statistics; what carried into Section 5 is the last column.

What carried over. Three results of the tour became design decisions rather than findings. The probing station failed in the informative direction: the probe and the difference of means were the same direction at every setting, because sentence pairs that differ in one word leave a probe nothing else to separate on, which is why the families in Section 5 vary the template instead of a single word. The SAE station produced the method of comparison: reading two directions at equal effect instead of at equal dose turned a clear-looking loss into a tie, and that reading is what all four sources later get. The steering station produced the dose rule and the reason it is not optional, since a random vector of



the same norm buys most of the same damage while moving nothing. The patching station produced no ingredient, only the habit of not trusting a ranking that was never measured causally.

Which model. The tour covered Gemma-2-2B, and the experiment still runs on GPT-2 small. That was decided by a measurement rather than by preference. The anchor’s own scaling table reports 5 of 23 concepts steerable at 160M parameters, which is a floor effect large enough to make the whole comparison vacuous, so before committing I measured A_{lin} on three candidate families: 0.77, 0.91 and 0.29, each peaking at layer 10 and each exactly zero up to layer 7. Two families well above the anchor’s go/no-go threshold of 0.1 and one in between is enough spread to work with. GPT-2 small also has published sparse autoencoders for its residual stream [13], and the sweep in Section 5 uses one of them per layer. This is a budget decision, taken after a pre-measurement rather than before one.

4 Choosing the question

What and why. Between the tour and the experiment sits a decision: out of four methods and a large literature, one question had to be small enough to finish and specific enough to come out wrong. This section says how I surveyed the field, which question I picked and why, which paper I use as an anchor, and which two gaps in it became the measurement in Section 5.

The map. Before choosing I surveyed the four methods of the tour and wrote one row per method: what the field measures, what counts as settled, and which gap is reachable on a laptop. The protocol was fixed before the first search: arXiv and the main proceedings, emphasis on work since 2024, a paper is only included if it reports its own measurement on open weights of at most nine billion parameters, and every entry carries an evidence level recording whether I read the full text, the abstract, or a secondary description. The last rule exists because the search was assisted by a language model, and fabricated references are the systematic failure mode of that: each arXiv identifier was resolved against the arXiv API and checked against title and first author. About 160 titles were screened, 55 arXiv entries kept.

What stood out. All four methods have the same core question under different names: the corrupt baseline in patching, the KL filter and the integer dose grid in steering, the missing control task in probing [9], [10], the sparsity level in autoencoders [12]. And the gaps reachable without training anything are control and baseline work, not new methods.

The cut. Of the candidates that came out of the map I chose the one that asks whether a direction which **reads** a concept well also **steers** it well, measured across concepts, layers and sources of the direction. Three reasons. It sits on the core question above, because reading and steering are two instruments pointed at the same concept, and the interesting case is the one where they disagree. It needs no tooling I did not already have, since probes, difference-of-means directions, SAE latents and the steering loop all exist from the tour. And it has an anchor: a recent paper with a central claim cheap enough to reproduce before extending it. The candidate I dropped last was a dense dose-response study with capability benchmarks, not because the question is uninteresting but because most of its work would be harness building. The control-task candidate fits inside this one as a step, so it was not dropped at all.

The anchor. Billa [14] proposes A_{lin} and reports that it predicts where steering will work: across 24 controlled binary families it correlates with steering effect at $\rho = +0.86$ to $+0.91$ and with the choice of layer at $+0.63$ to $+0.92$, while the customary “use a middle layer” heuristic lands on $A_{\text{lin}} = 0$ on Gemma-2-2B and does nothing there.



Its limits, by its own account. Much of that correlation comes from families sitting at the floor, and restricted to $A_{\text{lin}} > 0.1$ it drops to about +0.5 on the smaller models; the tasks are single-token throughout; and steering means a difference of means and nothing else.

The two gaps. Two things the paper does not settle were within my reach. It never states the dose at which it steers, and it reports $\rho(\|d\|, \text{KL}) = +0.96$, so its side-effect axis is very nearly just the norm of the vector it added. And its expectation that SAE features should help where a concept is encoded non-linearly is marked explicitly as untested.

The second work. Tiwari et al. [15] separate decodability from causality: on Gemma-2-9B-it the features geometrically closest to a probe overlap only 11 to 14% with the features that gradients mark as behaviourally relevant. They ablate rather than steer, have no difference-of-means source, and compare at equal set size rather than at equal effect, so what follows is not a rebuild of their study. What I take from them is one check: in their Qwen3-8B experiment 74% of the probe's margin lived in the reconstruction error of the SAE, and where that happens an SAE intervention mostly measures itself. I report that share per family rather than assume it away.

The contribution in one sentence: measure the anchor's diagnosis at equal effect and equal side effect instead of at whatever dose the norm of a difference of means happens to produce, and test its own untested prediction about SAE features while doing so.

Both gaps are cheap to close: a stated dose axis, a random control and two further sources of the direction, no new method.

5 Reading versus steering

What and why. This is the experiment the report is built around and the only part with results of its own. It starts from the anchor's claim [14] that A_{lin} , the logit-lens hit rate of Section 2, tells in advance where a steering vector will work. Section 5.1 gives the setup and Section 5.2 reproduces the claim on a model more than an order of magnitude smaller than the ones it was shown on. Section 5.3 is the part I added: the anchor never says how much it adds, so I compare four sources of a steering direction at equal side effect instead of at whatever dose each one happens to have. Section 5.4 says what survives that comparison, Section 5.5 what this setup cannot decide. If you read one table, read Table 5.

5.1 Setup

The families follow the anchor's design: two answer classes, a shared sentence template, and a single-token answer, which matters because it makes the effect of an intervention a change in one probability, with no decoding choices entering the measurement. I built six families and picked them by measuring A_{lin} beforehand (Table 3). Four carry signal on GPT-2 small; two do not, and they stay in the set for exactly that reason: if a method reports an effect on parity, the method is wrong, not the model. The intervention always pushes the second class towards the first class's answer, so the direction keeps one sign throughout.

Where possible a family uses several sentence templates, because with a single one a probe can separate the classes on the subject word alone. continent and parity have one each, and their probe numbers should be read with that in mind.

Family	Answer A / B	Prompts	Templates	Peak A_{lin}
pronoun	she / he	36 + 36	3	0.99
temperature	cold / hot	36 + 36	3	0.68
daynight	day / night	16 + 16	16	0.31
size	small / big	36 + 36	3	0.25
continent	Europe / Asia	12 + 12	1	0.04
parity	odd / even	18 + 18	1	0.00

Table 3: The six families, ordered by peak A_{lin} . Answers are single tokens and carry the leading space GPT-2 tokenises them with. The last two are floor cases: the model cannot answer them at all, so nothing should be steerable there.

Three ways of reading a concept off a layer are measured per layer:

- A_{lin} — the model’s own unembedding applied to the residual stream after each layer, argmax compared with the answer token, share correct. This is the logit lens [4] used as a predictor rather than as a description; it needs no training and one forward pass.
- **A logistic probe** on the same residual stream, fitted per layer, scored on a held-out quarter, with a control task on shuffled labels beside it [8], [9].
- **AUROC of the difference-of-means direction**: how well the single direction between the class means separates held-out examples.

And one way of writing one in: add a direction at every prompt position of the second class’s prompts. Its **effect** is ΔP , the change in probability of the first class’s answer token; its **side effect** is the Kullback–Leibler divergence of the next-token distribution on 50 unrelated prompts. Both are the anchor’s measures, so its numbers and mine are comparable. KL on unrelated prompts is a narrow notion of damage, since it says nothing about generation quality, but it is cheap, it means the same thing for every family, and it needs no judge.

Everything runs on GPT-2 small [16] (124M parameters, 12 layers, $d_{\text{model}} = 768$) through Transformer-Lens [17], seed 0, on a laptop GPU. The two notebooks run top to bottom in about 20 seconds and about three and a half minutes and write their numbers to results/.

5.2 The anchor, reproduced

The anchor reports its correlations on five models from Pythia-2.8B to Llama-8B. GPT-2 small is smaller than all of them, so reproducing the pattern here is a test rather than a confirmation: a measure like this could be an artifact of scale. It is not. For every family except pronoun, A_{lin} is exactly zero in layers 0 to 7 and rises only from layer 8 on, the same shape the anchor reports in layers 18 to 24 of a 2B model. pronoun is the exception and jumps to 0.64 at layer 6. Within families, $\rho(A_{\text{lin}}, \Delta P)$ across the twelve layers is +0.48 to +0.90, inside the anchor’s band of +0.63 to +0.92.

The probe fails as a layer selector, and “worse than A_{lin} ” understates how. For continent and pronoun the held-out probe reads 1.00 in every one of the twelve layers: it has no variance left to rank layers with, so its correlation with steering effect is not weak, it is undefined. Where it is defined it changes sign by family, +0.74 on daynight against −0.39 on size.

continent is the sharpest case, and the one the subtitle refers to:

On continent the probe reads the concept perfectly in all twelve layers, against a control task at 0.17 to 0.67, so this is not the probe memorising labels. A_{lin} peaks at 0.042, and steering moves the answer by at most +0.014. The concept is present, linearly readable, and unsteerable.

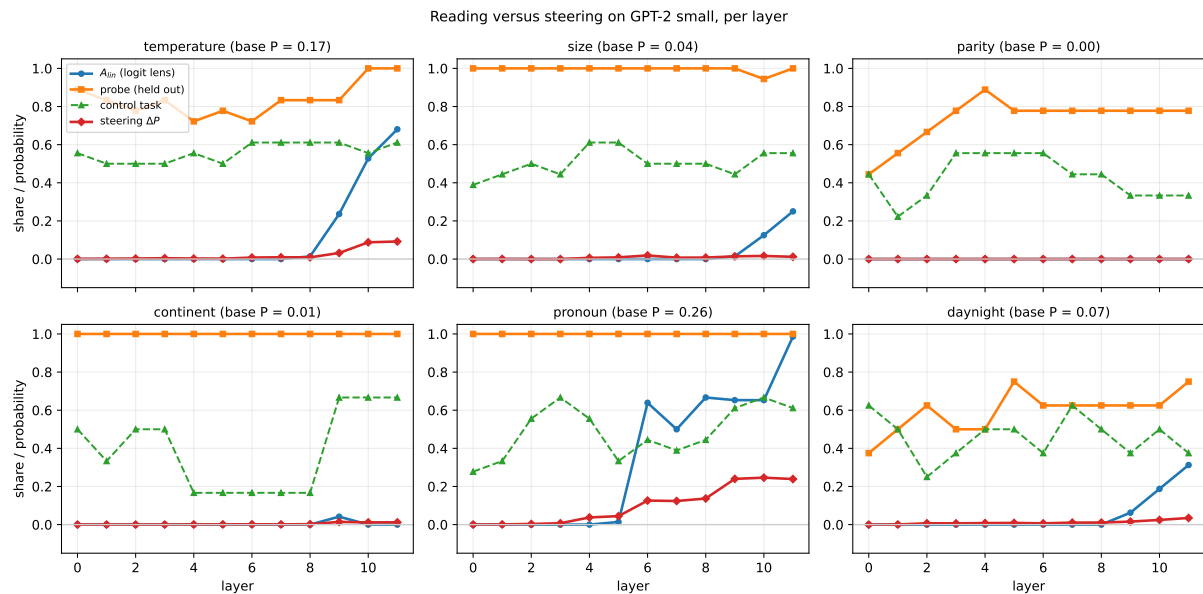


Figure 2: Reading against steering per layer, one panel per family. A_{lin} (circles) rises late; the probe (squares) is flat and high wherever the family has one template; the control task (triangles) shows the probe is not merely memorising; steering ΔP (diamonds) follows A_{lin} , not the probe. Source: `notebooks/07_reading_vs_steering.ipynb`.

Two further claims of the anchor hold. The common “use a middle layer” heuristic loses exactly where A_{lin} says it should: temperature gains $+0.092$ at layer 11 against $+0.008$ at layer 6, and daynight $+0.035$ against $+0.007$. And the go/no-go threshold survives: parity and continent stay below 0.05 peak A_{lin} and produce $\Delta P \leq +0.014$.

Then the dose. The anchor adds the raw difference of means at coefficient 1, and across these six families and twelve layers that vector’s norm ranges from 0.5 to 58.4, while the residual norm grows by a factor of eight across depth (Section 2). Coefficient 1 is therefore a different dose in each of the 72 cells, and an efficiency measure of $\Delta P / \text{KL}$ compares concepts at whatever dose their difference of means happens to have.

5.3 Four sources of a direction, at equal side effect

What and why. Same six families, same seed, same 50 unrelated prompts; three things change. The dose is stated and held fixed relative to the layer. Four sources of the direction are compared instead of one, and one of them is random. And the comparison is read at equal side effect, because a direction that buys its effect with more damage is not a better direction. All of it serves one number: how much of the anchor’s reported correlation is its dose tracking its own predictor.

Dose. All four directions are scaled to unit norm and added as α times the mean residual norm of that layer. The whole comparison rests on this choice. The residual stream of GPT-2 small grows by a factor of 8.5 across depth ($76 \rightarrow 644$, then down to 489 in layer 11), so one absolute dose is a heavy intervention early and a light one late; a relative dose makes “the same amount” mean the same thing in every layer.

Sources. The difference of means; the probe’s weight vector; one SAE decoder row; and a random direction as the floor. The random one is the best of five draws in the layer scan and of three per dose point in the sweeps, not a single draw: one random vector is a sample, and a floor built from a sample moves. The SAEs are the public GPT-2 residual-stream set [13] (24 576 features per layer, $32 \times$

expansion), and they exist for layers 0 to 10.¹ Layer 11 has none, and layer 11 is where A_{lin} peaks for all four carrying families, so the SAE arm is blind at the most interesting layer. That is a property of the available weights, not of the method.

The feature is selected on the training half with a criterion fixed before looking: fire on at least 60% of one class and at most 25% of the other, with the sign fixed to the class the intervention pushes towards. How often that fails belongs in the report, because the alternative is to keep searching until something is found. In layer 6 nothing passes for four of six families and the fallback is the largest mean gap, with held-out AUROC of 0.38 to 0.79 – two of them below chance. In layer 10 the criterion is met everywhere except parity, with held-out AUROC of 0.88 to 1.00 for five families and daynight the exception at 0.40.

Readouts. A twelve-layer scan at fixed $\alpha = 0.1$, then a ten-point dose sweep (α from 0.01 to 0.4) in layer 6, the middle-layer heuristic, and in layer 10, the deepest layer with an SAE. Each sweep is read at one point fixed in advance: ΔP at $\text{KL}^* = 0.1$ nats, interpolated in log KL because KL is monotone in dose. ΔP itself is not: past some dose the effect collapses as the model stops producing a sensible distribution at all, so a comparison at a fixed dose can land on either side of that collapse. Figure 3 shows the three ways of reading the same two curves, and why only the third compares at equal cost.

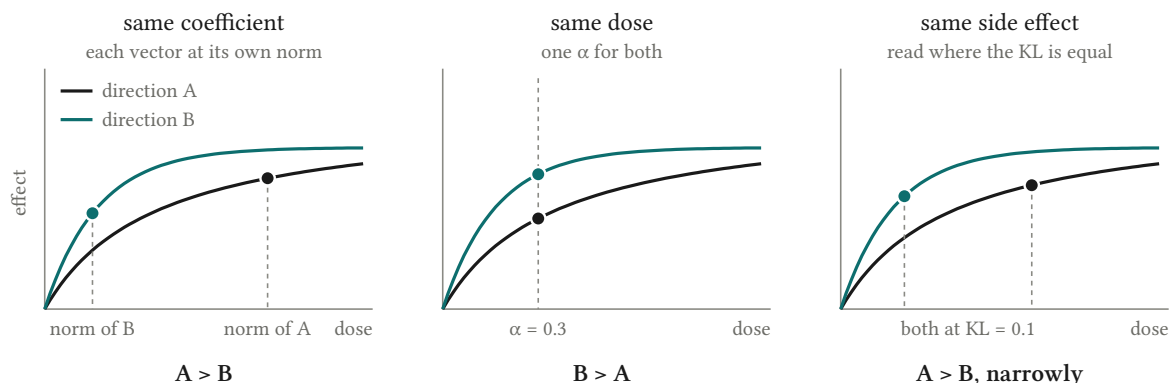


Figure 3: Own figure, schematic. The same two directions read three ways. At the anchor’s coefficient 1 each vector is added at its own norm, so the two are read at different doses; at a fixed relative dose both get the same amount; at equal side effect each is read where its KL reaches the budget. The curves are illustrative; the pattern is the one Table 5 and Table 4 show for the real families.

Family	Layer	diff-of-means	probe	SAE	random
pronoun	6	+0.263	+0.260	+0.245	+0.008
pronoun	10	+0.306	+0.307	+0.285	+0.011
temperature	6	+0.023	+0.109	+0.014	+0.015
temperature	10	+0.130	+0.156	+0.175	+0.017
size	6	+0.075	+0.022	+0.002	+0.039
size	10	+0.015	−0.015	−0.028	−0.013
daynight	6	+0.021	+0.005	−0.038	+0.051
daynight	10	+0.040	+0.032	+0.023	+0.024

Table 4: Effect ΔP at the same side effect ($\text{KL} = 0.1$ nats), best per row in bold. The two floor families are omitted: parity is exactly 0.000 in every cell, continent at most +0.023.

¹The folder names are off by one: `blocks.{l+1}.hook_resid_pre` is bit-identical to `blocks.{l}.hook_resid_post`, so folder $l + 1$ holds the SAE for layer l . I checked this instead of trusting the names.

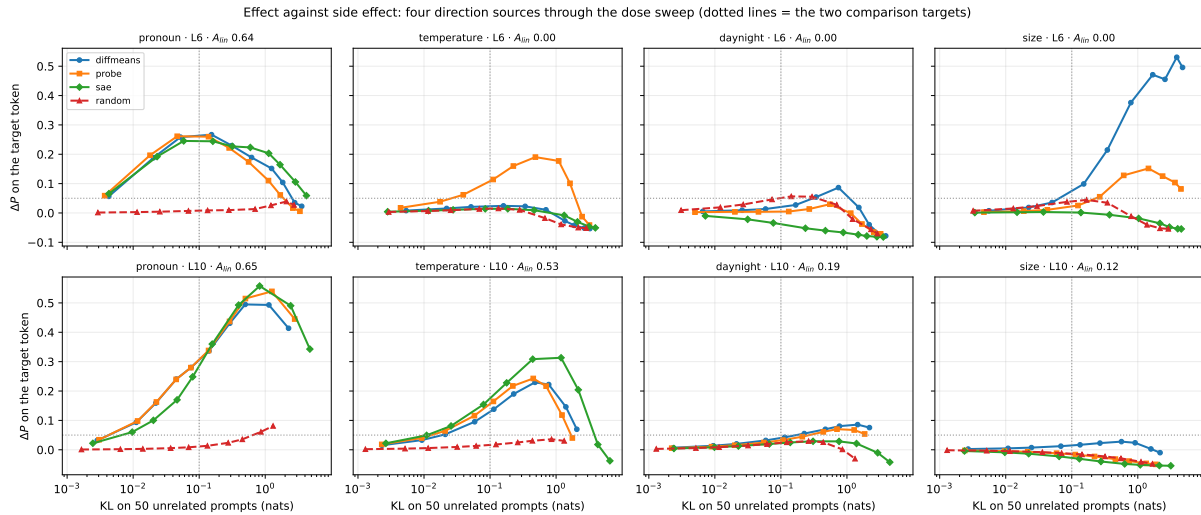


Figure 4: Effect against side effect for the four sources, top row layer 6, bottom row layer 10, log KL on the x-axis. The dotted lines are the two comparison targets. Each curve is one direction through ten doses; the random floor (dashed) is the best of three draws per point. Source: notebooks/08_direction_sources.ipynb.

5.4 What survives the fairer comparison

The anchor’s implicit dose, the norm of its difference of means divided by the residual norm of the layer, ranges from 0.003 to 0.119 across families and depth, and within each carrying family it correlates with A_{lin} at +0.76 to +0.91 (Table 5). Figure 5 shows what that number means: the dose rises in the same layers as the predictor. Holding the dose fixed, the within-family correlation between A_{lin} and steering effect drops in every family, in one of them below zero (last two columns of Table 5).

Family	Peak A_{lin}	Implicit dose	$\rho(A_{lin}, \text{dose})$	ρ at coeff. 1	ρ at $\alpha = 0.1$
temperature	0.68	0.004 – 0.085	+0.84	+0.80	+0.56
size	0.25	0.006 – 0.117	+0.76	+0.57	−0.10
pronoun	0.99	0.005 – 0.086	+0.91	+0.90	+0.54
daynight	0.31	0.005 – 0.067	+0.76	+0.76	+0.24
continent	0.04	0.005 – 0.119	+0.31	+0.48	−0.22
parity	0.00	0.003 – 0.013	flat	flat	flat

Table 5: The dose confound, in numbers. Column 3 is the dose the anchor applies without stating it; column 4 shows it tracking the anchor’s own predictor. Columns 5 and 6 are the same correlation at the anchor’s dose and at a fixed relative dose. parity has $A_{lin} = 0$ in every layer, so there is no rank order to correlate.

That looks like a refutation, and my first reading of it was exactly that: at a fixed dose the ranking of the families does not merely weaken, it inverts, with size moving from last place to first. I held that reading back for one reason. At a fixed dose the four families do not pay the same side effect, so the fixed-dose ranking is no fairer than the anchor’s (Figure 3, middle panel). Read at an equal KL budget instead, the order comes out as pronoun > temperature > size > daynight against the predicted pronoun > temperature > daynight > size, one adjacent swap.

The apparent collapse of the ranking is itself a dose artifact. The weakening of the correlation is not. The anchor’s predictor is real and its reported strength is inflated by a dose it never states.

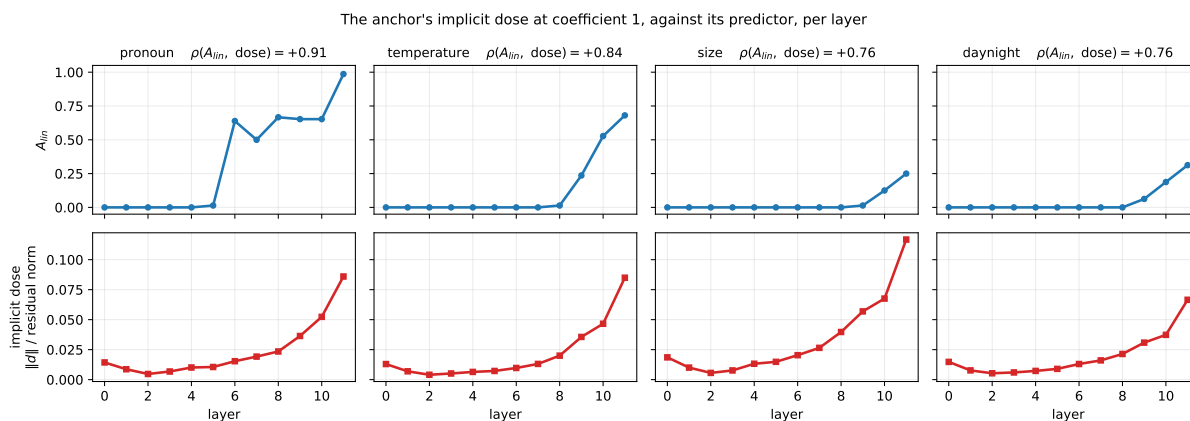


Figure 5: The dose confound per layer, for the four carrying families. Top: A_{lin} . Bottom: the dose the anchor applies at coefficient 1, the norm of the difference of means divided by the mean residual norm of the layer. Both rise from layer 8 on; the ρ in each panel title is the fourth column of Table 5. Drawn from `results/` by `scripts/fig_dose_confound.py`.

Four further findings, all from Figure 4 and Table 4.

The unstated dose is also not the best one. With the same direction in layer 10, pronoun reaches $\Delta P = +0.495$ at $\alpha = 0.2$, against $+0.246$ at the anchor's coefficient 1.

Where the direction comes from matters less than which layer and how much. In layer 10 the three principled sources land within 0.03 of each other on pronoun and within 0.05 on temperature. The large differences in the grid are between layers, not between derivations, which is awkward for a literature that mostly compares methods of extraction.

The anchor's untested SAE prediction does not hold uniformly. Its third regime (high A_{lin} , so the concept is already output-aligned and an SAE feature should add little) holds for pronoun ($+0.285$ against $+0.306$) and fails for temperature in layer 10, where the SAE direction wins outright ($+0.175$ against $+0.130$).

Explained variance is blind to the concept. On the four carrying families the SAE reconstruction explains 93 to 99% of the variance of the residual stream, while 35 to 99% of the class separation along the reading direction sits in the reconstruction error, the share Tiwari et al. measure at 74% on another model, another SAE and other concepts [15]. Replacing one layer by its own reconstruction costs the model its answer: accuracy on temperature falls from 0.68 to 0.35, on size from 0.25 to 0.07, on daynight from 0.31 to 0.09 (pronoun holds, 0.99 to 0.86). The metric measures reconstruction, and reconstruction is not the concept.

Two things the controls caught. The random floor is not zero and it wins once: on daynight in layer 6 it beats all three principled sources at equal side effect ($+0.051$ against $+0.021$, $+0.005$ and -0.038). And parity stays at exactly 0.000 across four sources, twelve layers and ten doses; a pipeline reporting an effect there would be reporting its own noise. [18] find the dissociation of Section 5.2 from the other side on a larger model, where a perfectly separating detection direction stands at 83° to the direction that actually steers.

5.5 What this setup cannot decide

The directions are fitted on the same prompts that are then steered, as in the anchor; there is no held-out steering set, so the effect sizes are upper bounds. The families have 24 to 72 examples against 768 dimensions, so the held-out quarter is 4 to 18 prompts, which is why no probe number here is quoted to three digits. The cross-family correlation rests on six points and is an illustration, not evidence; the



anchor uses 24. I also left out the anchor's second measure, A_{mlp} , so its middle regime (present but not linearly readable) rests on the probe alone here, and a failing probe never proves absence [10]. The random floor is the best of a few draws, not a distribution, and the SAE arm cannot see layer 11, where four of six families peak.

None of this is fatal to the conclusion, which is a comparison between directions measured the same way rather than an absolute claim about steerability. It does mean that every number in this section is a statement about GPT-2 small with these six families; the anchor's own result across five larger models is the reason to think the pattern is more than that.

6 Conclusion

The question was whether a training-free predictor beats a trained one at saying where steering will work, and whether it matters where the steering direction comes from. The probe cannot answer the first one: it saturates at 1.00 across all twelve layers on two families, which leaves nothing to correlate, while A_{lin} ranks the families in the order their steering effects arrive. The anchor's correlation holds and is overstated: the dose it never states tracks its own predictor at +0.76 to +0.91, and at a fixed dose its ranking inverts. Read at an equal side-effect budget instead, which is the comparison I would defend, the predicted order survives with one adjacent swap. On the second question the answer is that the derivation matters least: in layer 10 the difference of means, the probe and the sparse-autoencoder feature land within 0.03 to 0.05 of one another, while the same source read in layer 6 instead of layer 10 moves temperature from +0.023 to +0.130. The controls decided what all of it means: a random vector of the same norm won one cell outright, and the family the model cannot answer stayed at exactly 0.000 everywhere.

Two questions remain, and I would put them to any steering result including my own: at what dose, and compared at equal what?

References

- [1] L. H. Gilpin, D. Bau, B. Z. Yuan, A. Bajwa, M. Specter, and L. Kagal, "Explaining Explanations: An Overview of Interpretability of Machine Learning," in *IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA)*, 2018.
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why Should I Trust You?': Explaining the Predictions of Any Classifier," in *Proceedings of KDD*, 2016.
- [3] N. Elhage, N. Nanda, C. Olsson, and others, "A Mathematical Framework for Transformer Circuits," *Transformer Circuits Thread, Anthropic*, 2021.
- [4] nostalgebraist, "Interpreting GPT: The Logit Lens." 2020.
- [5] K. Park, Y. J. Choe, and V. Veitch, "The Linear Representation Hypothesis and the Geometry of Large Language Models," *arXiv preprint arXiv:2311.03658*, 2024.
- [6] S. Heimersheim and N. Nanda, "How to Use and Interpret Activation Patching," *arXiv preprint arXiv:2404.15255*, 2024.
- [7] F. Zhang and N. Nanda, "Towards Best Practices of Activation Patching in Language Models: Metrics and Methods," *arXiv preprint arXiv:2309.16042*, 2024.
- [8] G. Alain and Y. Bengio, "Understanding Intermediate Layers Using Linear Classifier Probes," *arXiv preprint arXiv:1610.01644*, 2017.



- [9] J. Hewitt and P. Liang, “Designing and Interpreting Probes with Control Tasks,” in *Proceedings of EMNLP-IJCNLP*, 2019.
- [10] Y. Belinkov, “Probing Classifiers: Promises, Shortcomings, and Advances,” *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022.
- [11] A. M. Turner *et al.*, “Steering Language Models with Activation Engineering,” *arXiv preprint arXiv:2308.10248*, 2024.
- [12] H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey, “Sparse Autoencoders Find Highly Interpretable Features in Language Models,” *arXiv preprint arXiv:2309.08600*, 2023.
- [13] J. Bloom, “Open Source Sparse Autoencoders for All Residual Stream Layers of GPT2-Small.” 2024.
- [14] J. Billa, “Predicting Where Steering Vectors Succeed,” *arXiv preprint arXiv:2604.15557*, 2026.
- [15] D. Tiwari, C. Davis, S. Sinha, T. Weaver, A. Shah, and M. Chaudhary, “Decodability is Not Causality: Dissociating Probe Readouts from Behavioral Drivers via SAE Decomposition,” *arXiv preprint arXiv:2609.18080*, 2026.
- [16] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language Models Are Unsupervised Multitask Learners,” technical report, 2019.
- [17] N. Nanda and J. Bloom, “TransformerLens.” 2022.
- [18] Galeone and others, “Perfect Detection, Failed Control,” *arXiv preprint arXiv:2606.24952*, 2026.



A Appendix

Everything measured for this report is in github.com/Henroxx/mechinterp_xai, at the tag `v1.0-submission` — the state this text describes. The paths that document how the work was done are listed under the abstract; the table below is the measurement.

Path in the repository	What is in it
<code>notebooks/01_explore ... 06_sae</code>	The exploration of Section 3, one notebook per station: orientation, patching, steering, probing, a second model, sparse autoencoders. Single prompts and samples around twenty throughout.
<code>notebooks/07_reading_vs_steering</code>	Section 5.2: A_{lin} , the probe, its control task, difference-of-means AUROC and steering ΔP for six families across twelve layers. Runs top to bottom in about 20 seconds.
<code>notebooks/08_direction_sources</code>	Section 5.3: four sources of a direction over a ten-point dose sweep in layers 6 and 10, with the random floor and the reconstruction checks. About three and a half minutes.
<code>results/*.json</code>	The numbers behind Table 3, Table 4, Table 5 and both figures, as the notebooks wrote them. Re-running for this report reproduced the files byte for byte.
<code>report/</code>	This document: <code>main.typ</code> , <code>refs.bib</code> , and <code>figures/</code> with the notebook figures as vector PDF and the two schematics as SVG. <code>scripts/fig_dose_confound.py</code> draws Figure 5 from <code>results/</code> .
<code>pyproject.toml</code> , <code>uv.lock</code>	The pinned environment — <code>uv sync</code> rebuilds it, <code>uv run</code> starts the notebooks. Seed 0 throughout; <code>scripts/</code> holds the smoke test and the numerics check the setup was verified with.